

SPANDAN SHAH

Computer Engineering Student | Cybersecurity Minor | AI/ML, Space-Tech & Secure Systems

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SIH 2024
Finalist

8.93 CGPA
Till 6th Sem

Patent + Paper
Under Review

Space + Security
AI/ML + Full Stack

PROFESSIONAL SUMMARY

Computer Engineering student with a Minor in Cybersecurity, focused on AI/ML, space-tech, cybersecurity and full-stack scientific systems. Hands-on experience building exoplanet light-curve analysis tools, satellite imagery workflows, secure communication systems, AI security projects, backend APIs and React dashboards. SIH 2024 Finalist with a secure session architecture currently under patent and research review.

TECHNICAL SKILLS

Programming: Python, JavaScript, C, Java	Backend & APIs: FastAPI, Node.js, Express.js, REST APIs, Postman
AI/ML & Data: Scikit-learn, PyTorch, Pandas, NumPy, Feature Engineering	Computer Vision: OpenCV, YOLO, OCR, Image Processing
Space/GIS Data: TESS Light Curves, Satellite Data, GeoPandas, DEM, SAR, WebGIS	Cybersecurity: AES-GCM, Replay Protection, WAF, Phishing Detection, Secure Session Design

FLAGSHIP PROJECTS

ExoTrace - Exoplanet Candidate Detection TESS, BLS, ExtraTrees, FastAPI, React - Built a noisy light-curve screening pipeline using cleaning, Savitzky-Golay detrending, BLS detection, phase folding and feature extraction. - Classified Planet Candidate / False Positive / Eclipsing Binary signals with candidate thresholding and priority ranking. - Achieved 90% full-batch accuracy and 98% candidate recall on a 150-sample pilot dataset.	GoblinWisp - Secure Session Protocol Secure Systems, Authentication, Replay Resistance - Designed a secure peer-to-peer session architecture for infrastructure-free IoT, mesh and embedded environments. - Focused on identity-bound authentication, forward-secure sessions, directional isolation, counter validation and reset-safe handling. - Currently under patent/research review.
Vigilant Edge - AI Web Application Firewall Cybersecurity, AI, Web/API Security - Developing an AI-driven WAF for real-time web/API threat detection and traffic protection. - Implemented anomaly-focused detection for suspicious traffic, unknown attacks and high-risk request behavior. - Designed automated alerting, risk scoring and adaptive security response workflows.	AI/ML Satellite Imagery Change Detection BAH, ISRO, Satellite Data, GIS - Worked on automated feature detection and change analysis of glacial lakes, roads and urban drainage systems. - Explored optical, SAR and DEM processing for geospatial feature extraction and environmental monitoring. - Built the direction toward scalable environmental intelligence workflows.

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AI/ML | Cybersecurity | Space-Tech | Full-Stack Development

ADDITIONAL ENGINEERING PROJECTS

Berlin - Intelligent Traffic Surveillance YOLO, DeepSORT, OCR, OpenCV Developed vehicle detection, tracking, license plate recognition, speed estimation and CCTV traffic analytics.	PhishBuster - URL Threat Quantification AI, Heuristics, Cybersecurity Built phishing detection using URL heuristics, domain indicators, SSL checks, entropy analysis and risk scoring.
Question Paper Generator API Node.js, Gemini AI, REST APIs Created a Node.js API for automated exam paper generation using topic, question type, marks and exam pattern.	Wireless Communication Analysis ESP32, NRF24L01, BLE, Wi-Fi Studied BLE/Wi-Fi behavior, channel usage, device discovery, interference scenarios and SPI wireless data handling.

HACKATHONS, LEADERSHIP & PRACTICAL EXPERIENCE

SIH 2025 Student Coordinator Organized the college-level internal hackathon round; coordinated 20+ teams, mentors and faculty; managed engagement, allocation and evaluation flow. SIH 2024 Finalist / NCB Problem Statement Built a solution concept to identify suspicious users across Telegram, WhatsApp and Instagram using digital footprints and behavioral patterns.	SIH 2025 Developer (SIH25108) Built an AI-powered WebGIS decision-support concept for FRA monitoring and multi-state geospatial analysis. Organizer - Blackout Community Co-founded and managed a tech community focused on hackathons, project building, team collaboration and innovation guidance.
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RESEARCH, ACHIEVEMENTS & EDUCATION

PATENT / RESEARCH Patent - 1 under review Secure session architecture for infrastructure-free peer communication systems. Research Paper - 1 under review Based on the same secure systems research direction.	ACHIEVEMENTS SIH 2024 Finalist Selected as finalist in Smart India Hackathon 2024. Hackathons SIH 2024/2025, BAH 2025/2026, Volkswagen & IBM, Mobilothon 5.0, two Odoo hackathons.	EDUCATION B.E. Computer Engineering SAL Institute of Technology and Engineering Research 2023 - 2027 CGPA 8.93 till 6th Sem Class 12 / Class 10 80% / 85%
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